

E2, measured again.

A fresh smoke with all 20 graphs.

Fresh context construction, real Codex readers, deterministic scoring, and replayed evidence. This report shows what happened on the six-event smoke—not a configuration-selection result.

Infrastructure & tool replay: PASS

gpt-5.6-luna · medium

codex-cli 0.154.0

Fresh stores · no snapshot reuse

30/30

Valid, budgeted completions

All tool traces replay exactly

26/30

Exact answers correct

Five families, two questions each

24/30

Strict evidence checks

Scored separately from values

6

Fresh committed snapshots

Two windows per store

What changed in this run?

S3 answers the complete file-set question correctly this time, raising its value score from 9/10 to 10/10. S1 and S0 repeat their earlier value totals. The builders and readers are stochastic; the comparison does not isolate a code improvement.

CONTEXT STORE	EXACT VALUES	STRICT EVIDENCE	PREVIOUS VALUES	REPLAY AUDIT
S3 Codex-maintained indexed files	10/10	9/10	9/10	Pass
S1 Deterministic fixed template	7/10	7/10	7/10	Pass
S0 Raw event files	9/10	8/10	9/10	Pass

S3 · 10/10 VALUES

The file join succeeds this time

The fresh reader returns both changed file paths. Its current issue facts still mention only the newer PR, so success on ten questions does not prove complete fact preservation across the store.

S1 · 7/10 VALUES

Fixed-schema information loss persists

The template omits the release-freeze field. It cannot answer the two date questions and also abstains on the complete file-set question. A lossless deterministic baseline remains necessary.

EVIDENCE · SEPARATE METRIC

Correct abstention can fail citation policy

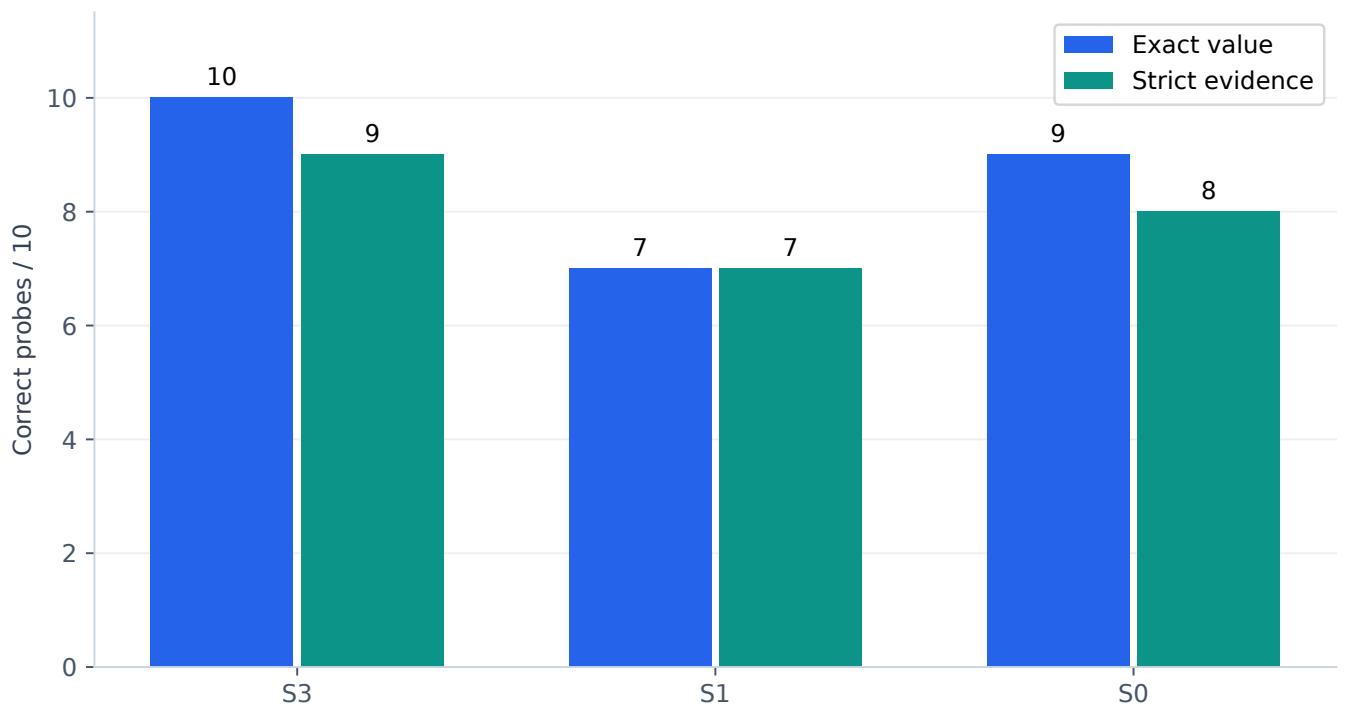
Several unknown answers cite events they read. The strict rule expects no citations for gold-unknown probes. Source-ID checks are not a general entailment judge; this contract needs freezing before the real-data study.

Interpretation limit: one fresh build per store and ten shared probes. No confidence intervals, statistical winner, semantic-retrieval comparison, graph comparison, or production-readiness claim follows from this smoke.

All measured smoke graphs

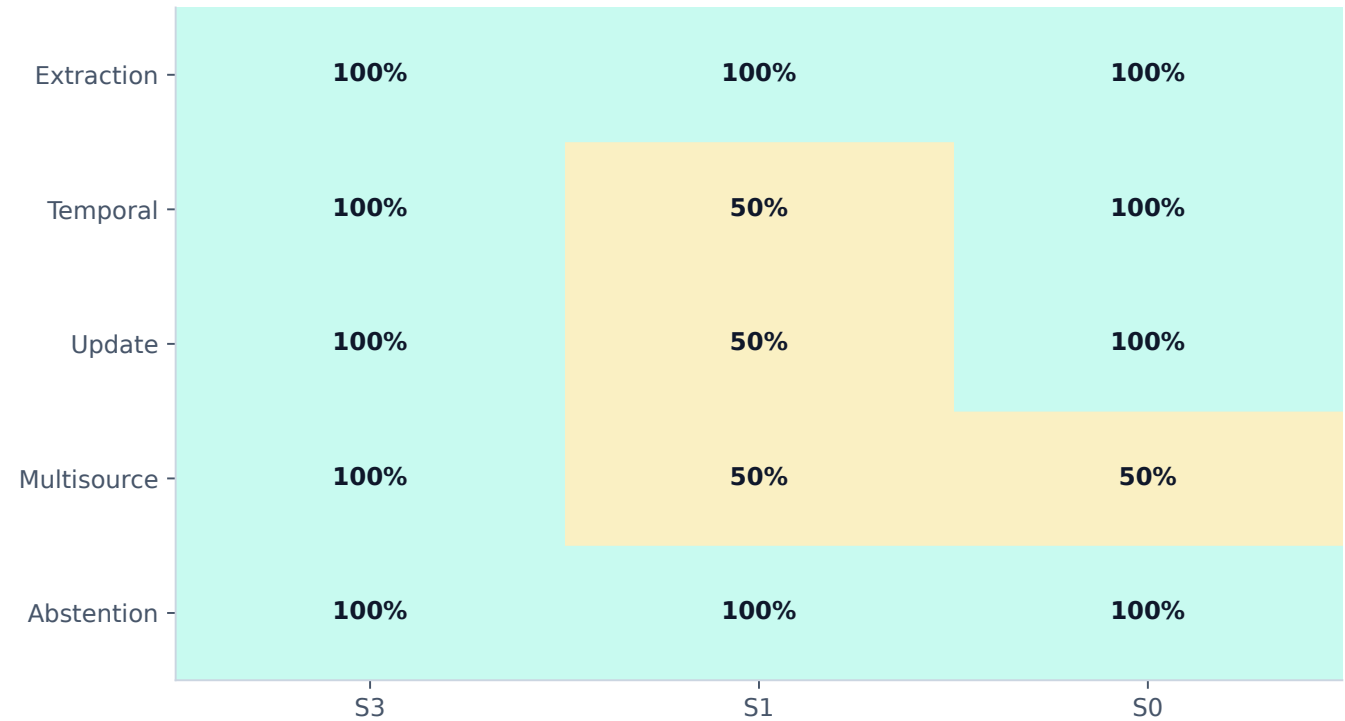
Every figure is generated from the new run's saved records. Filter by subject, click a graph to open it at full size, or download the SVG/PNG. Exact plot data is included in the export bundle.

Answer correctness and strict evidence



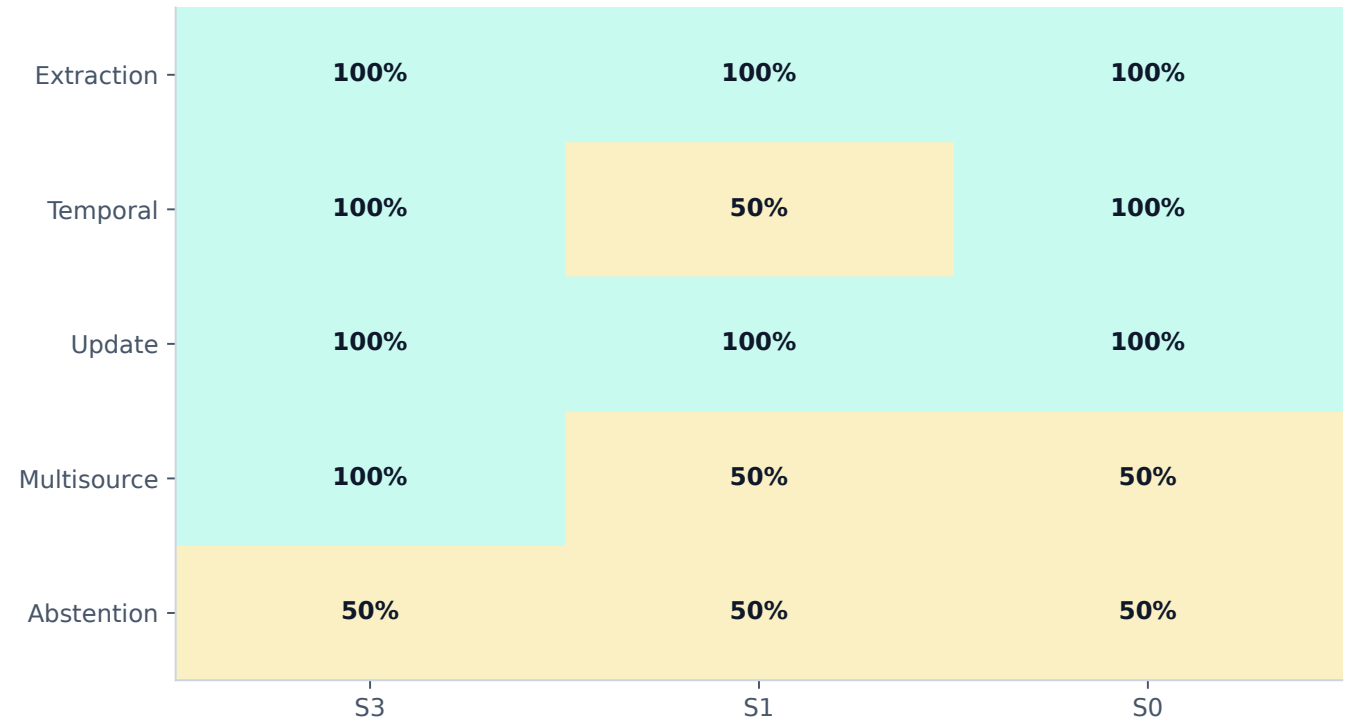
Each arm answers the same ten probes. Evidence checks required source IDs and actual retrieval exposure; it is separate from value correctness. No confidence intervals are estimated from this tiny smoke.

Exact answers by question family



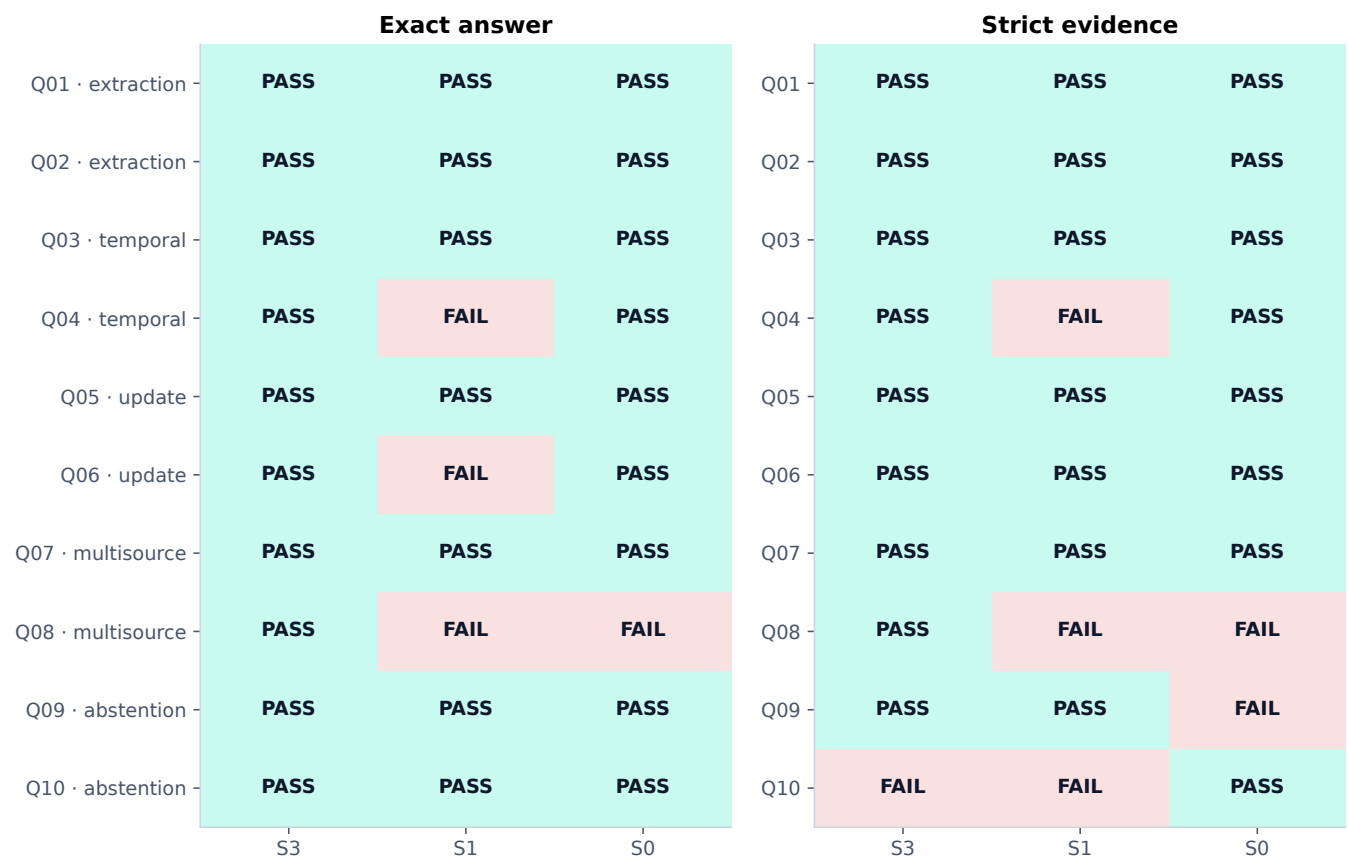
Two probes per cell: possible scores are 0%, 50%, and 100%. Equal family sizes make overall micro accuracy equal to the five-family macro mean in this fixture.

Strict evidence by question family



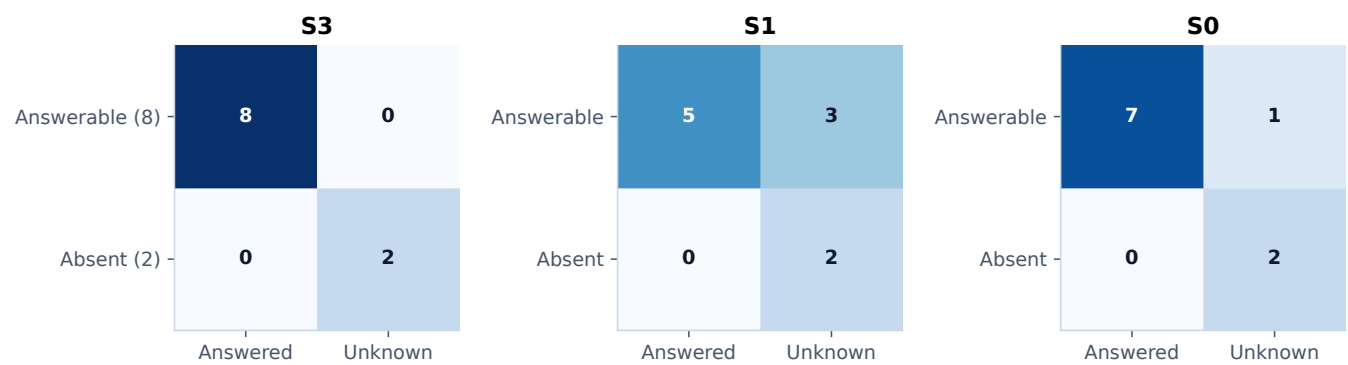
Two probes per cell: possible scores are 0%, 50%, and 100%. Equal family sizes make overall micro accuracy equal to the five-family macro mean in this fixture.

Every probe, every store



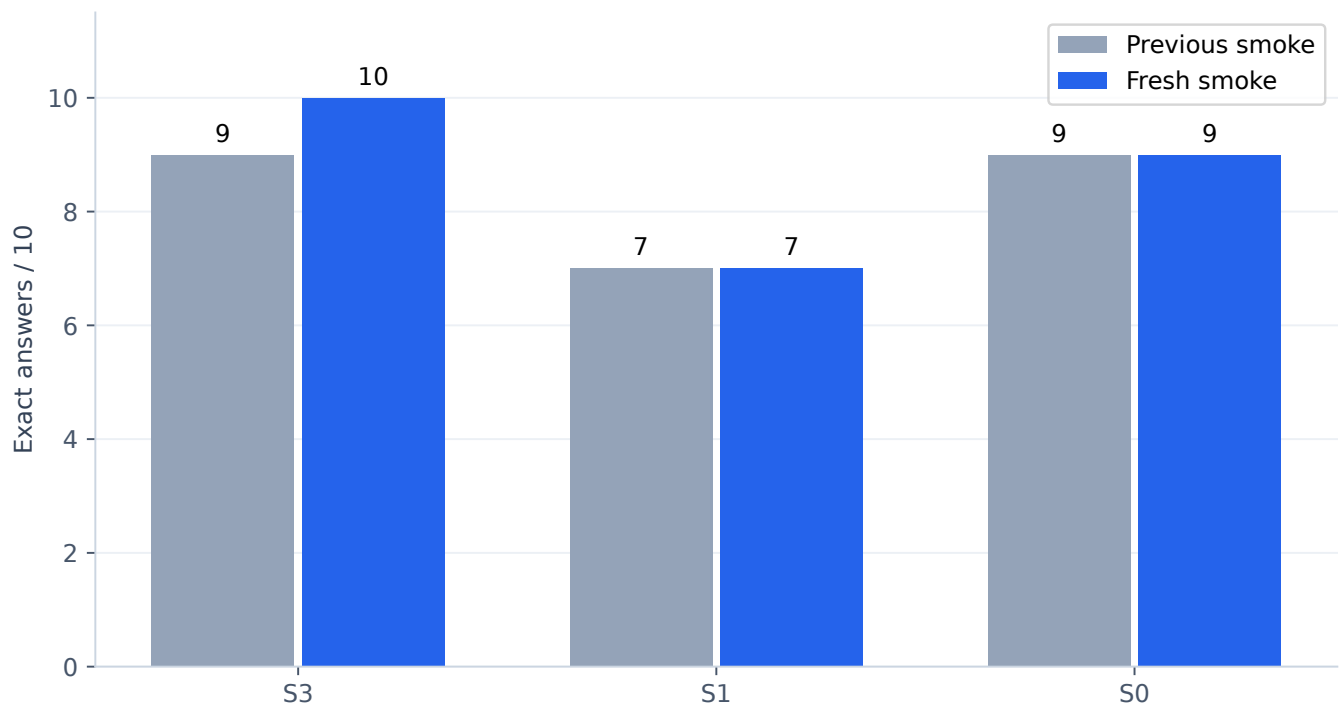
All 30 answers are retained, including wrong values and evidence failures. Q01–Q10 map to the expandable question table below.

Answer versus abstention decisions



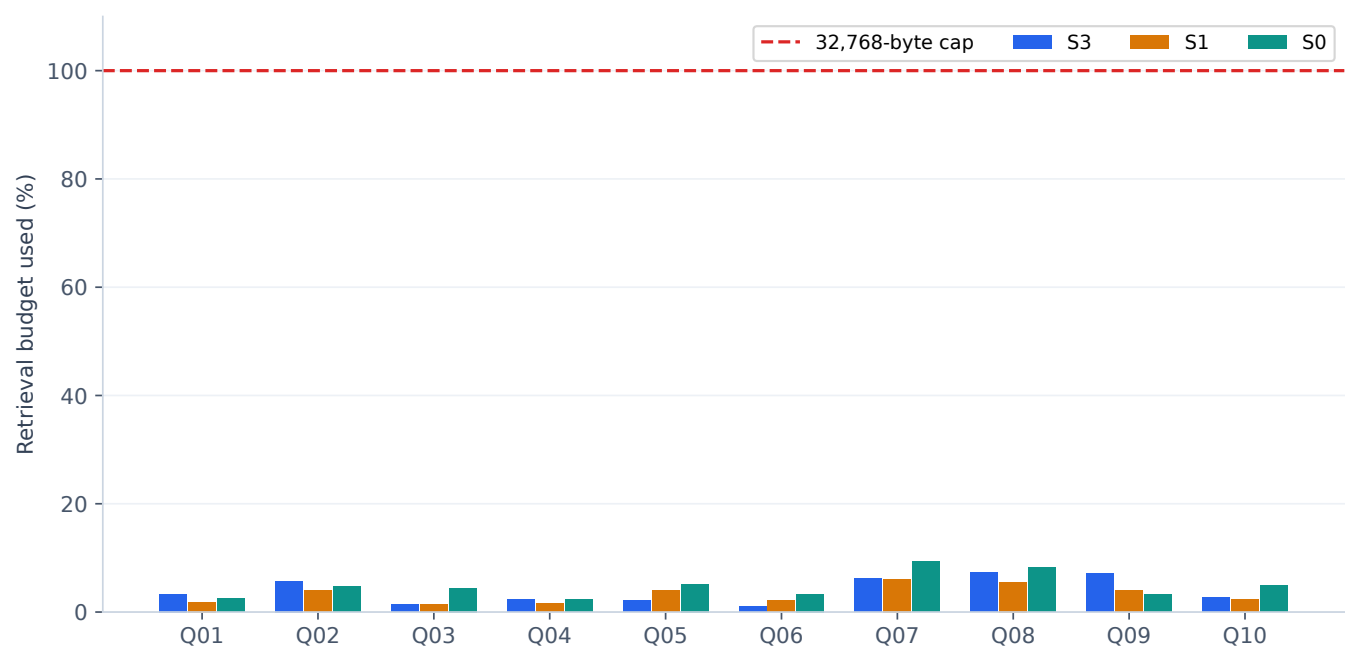
Rows are the gold answerability; columns are the model's decision. Answering an answerable question is not necessarily correct—the exact-value plots score that separately.

Fresh run versus the previous smoke



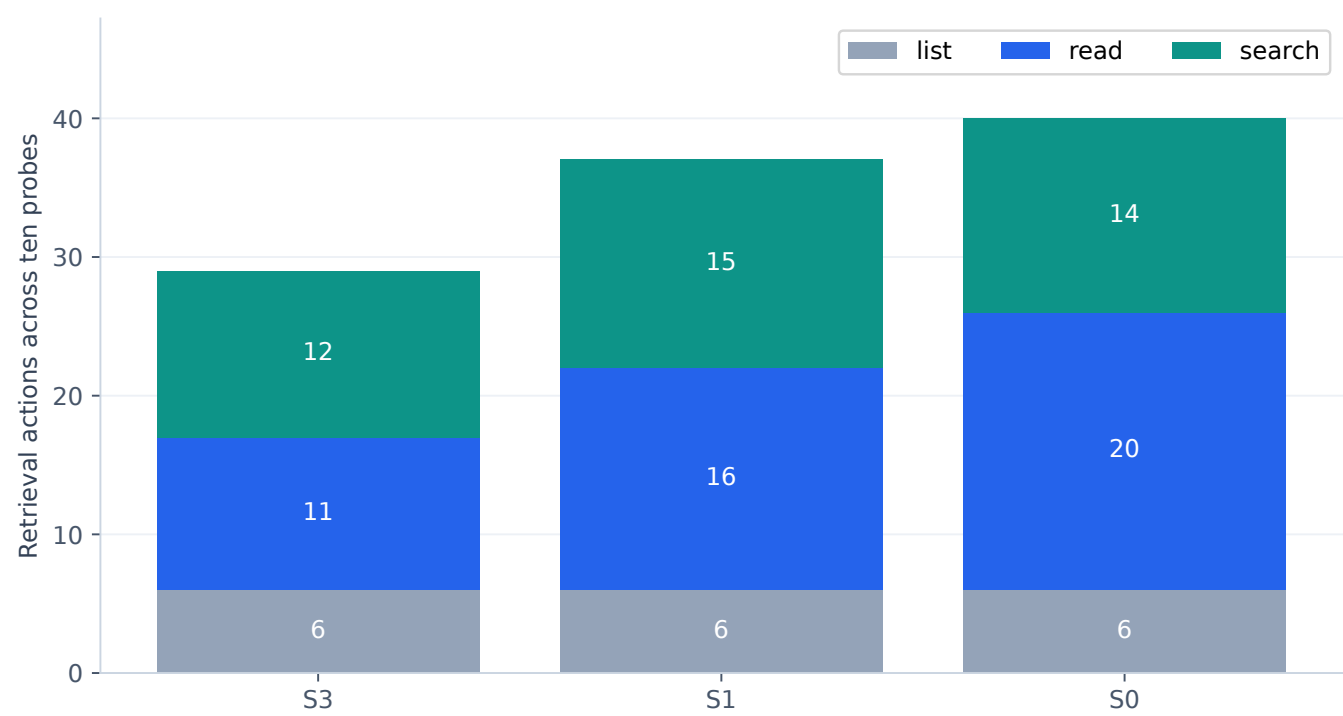
Same fixture and nominal model/effort; this run rebuilds the stores and resamples readers. Changes can come from either builder or reader stochasticity. Two runs do not establish a trend or a significant improvement.

Retrieval budget use for every probe



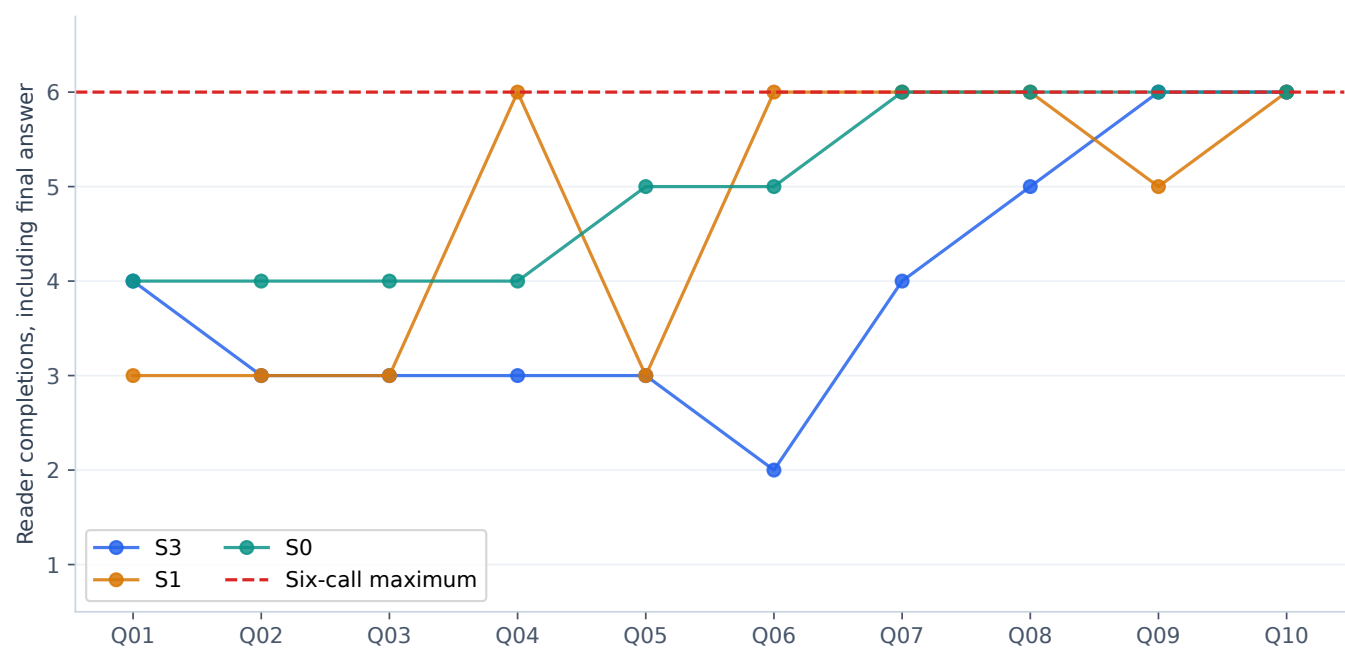
Every returned UTF-8 byte counts, including paths, snippets, errors and repeated reads. This is a byte budget, not a model-token cap. Unused budget does not imply that enough relevant evidence was retrieved.

Which tools did the reader use?



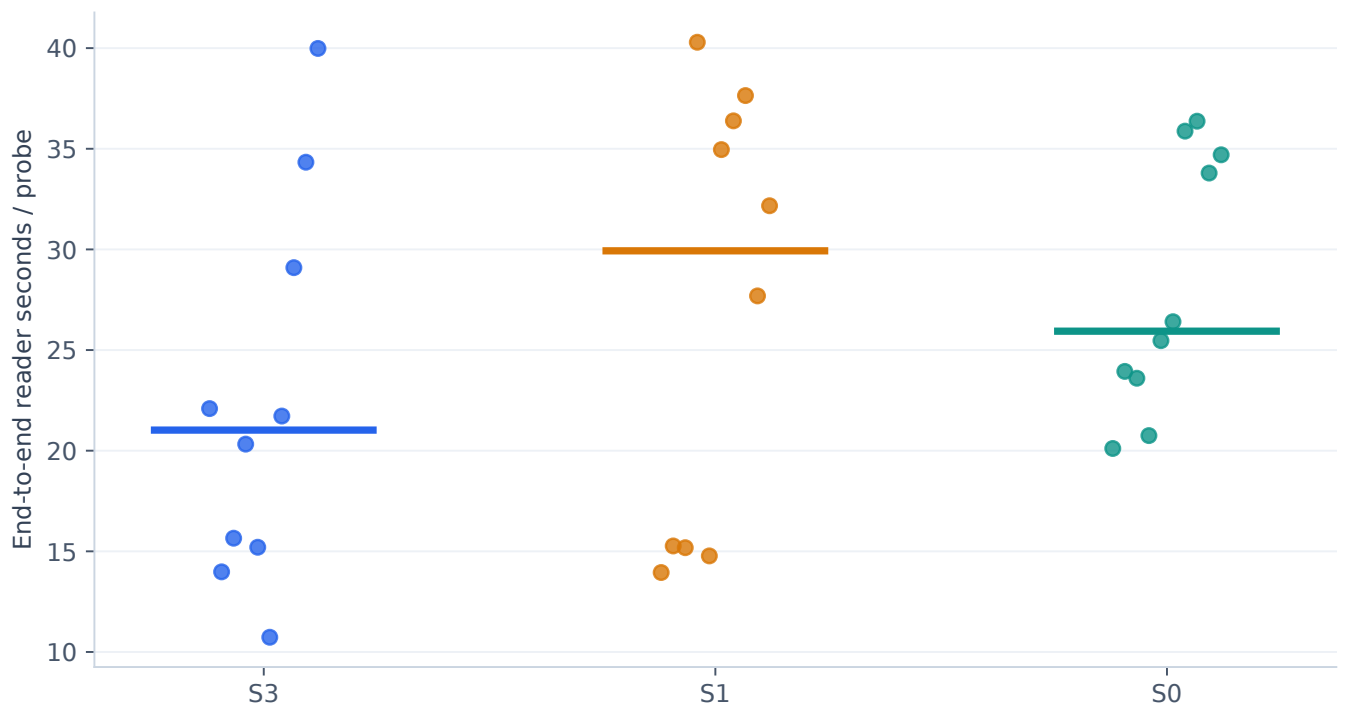
Host list/read/search actions only. The final answer and provider completions are not counted as retrieval actions. More tool calls alone are not evidence of better retrieval.

Reader call depth



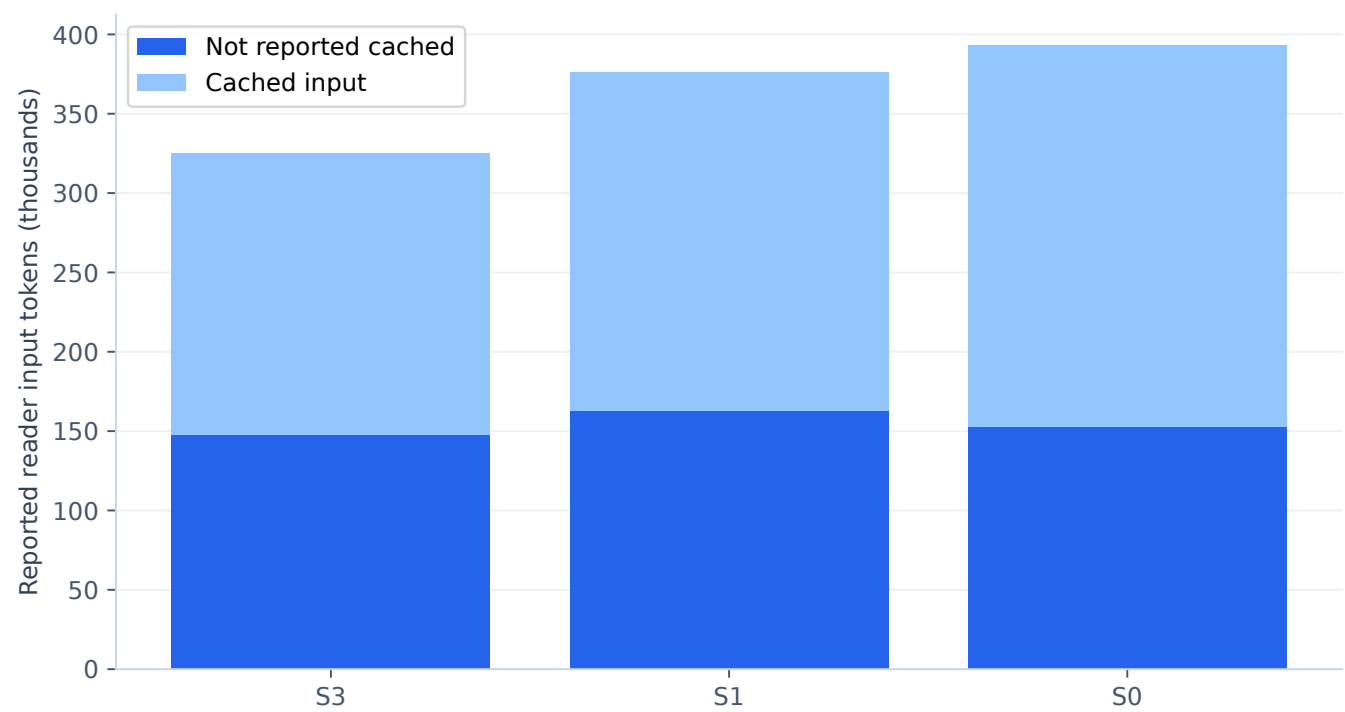
The sixth completion is reserved for an answer or unknown. A response at the cap may still be incomplete. Calls are measured independently of the retrieved-byte budget.

Observed reader latency



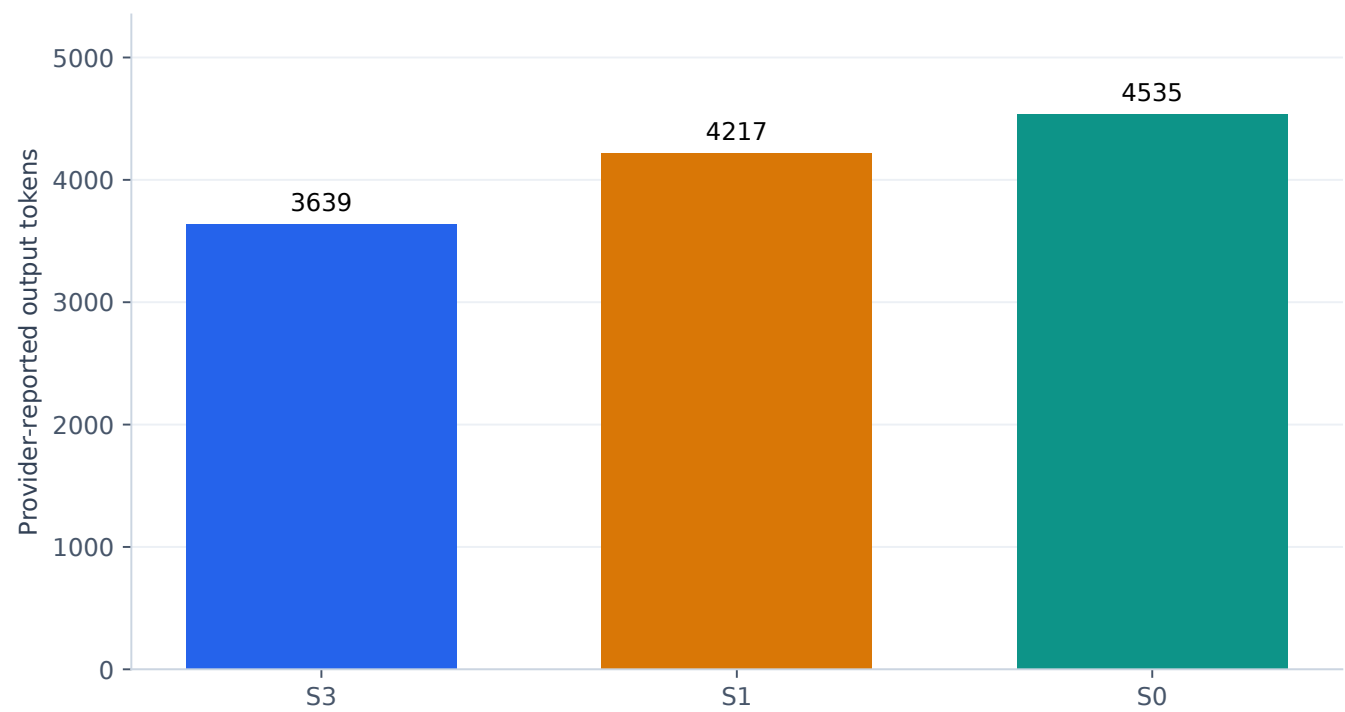
Dots are the ten probes; horizontal lines are medians. The three arm processes run concurrently. Values include repeated Codex startup/inference and local tool handling, and are not a controlled serial latency benchmark.

Reader input and cache accounting



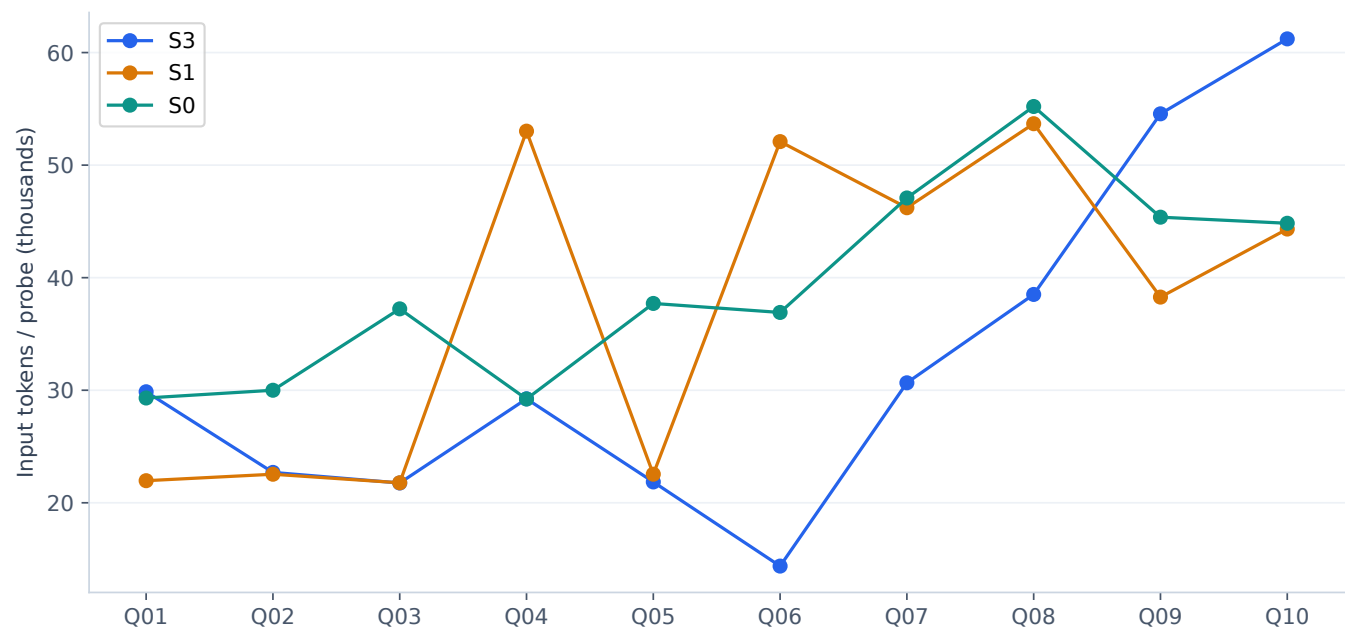
Cached input is a subset of total input, never added twice. Counts include instructions, repeated history and retrieved material across every completion. Token totals are not USD prices.

Reader generation volume



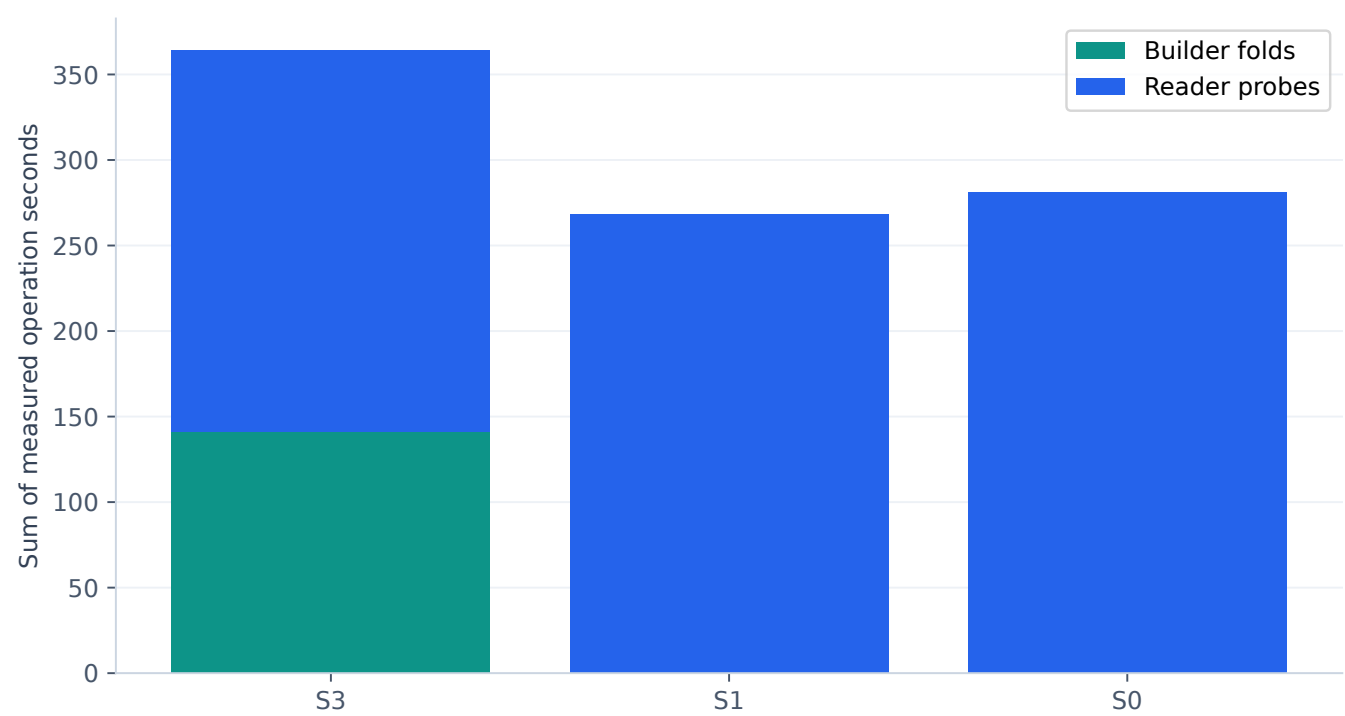
Reported output tokens across ten probes. Separately reported reasoning tokens are not added to this total. The CLI's output-token target is advisory; the enforced controls are calls, time and retrieval bytes.

Input footprint by question



Input is summed over all reader completions for that probe, including cache hits. High input can reflect repeated protocol overhead rather than more unique context read.

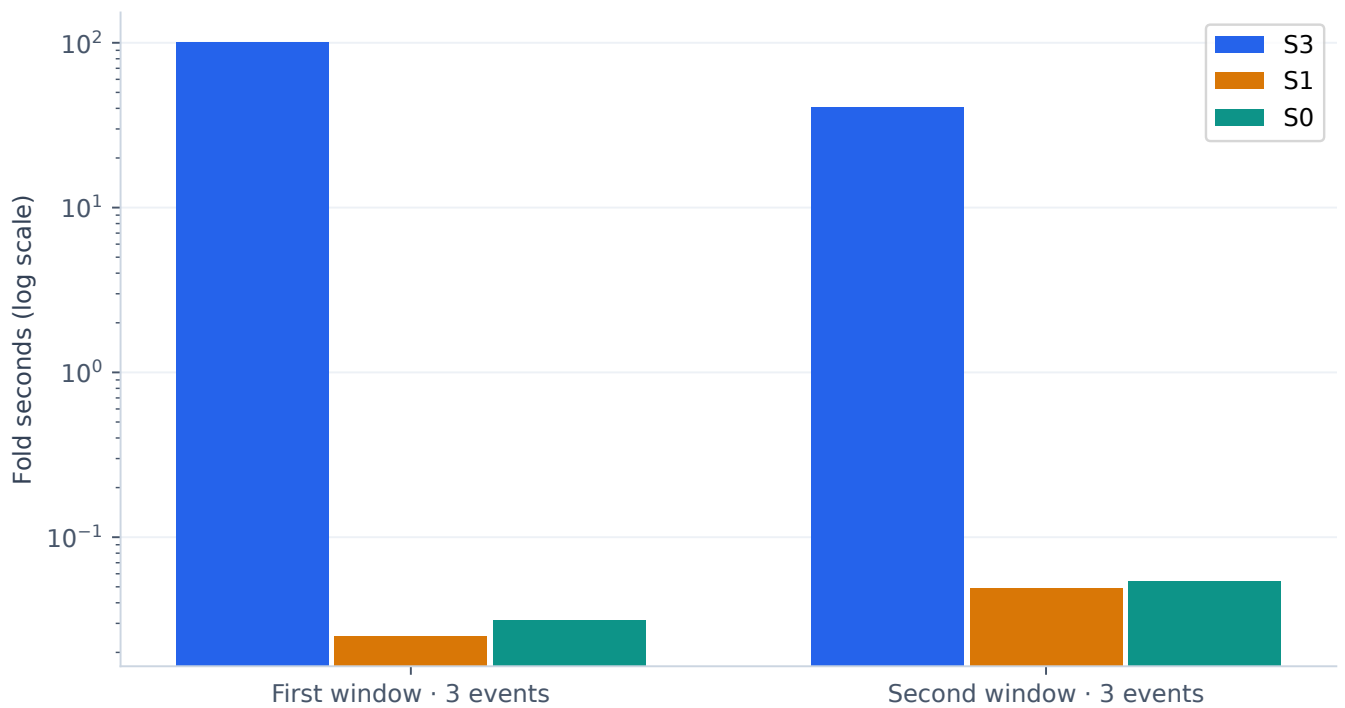
Measured work by phase



Within-arm builder and reader operation durations are summed. Setup/report overhead is excluded. Do not sum all three bars to estimate elapsed time: the arms run concurrently.

15

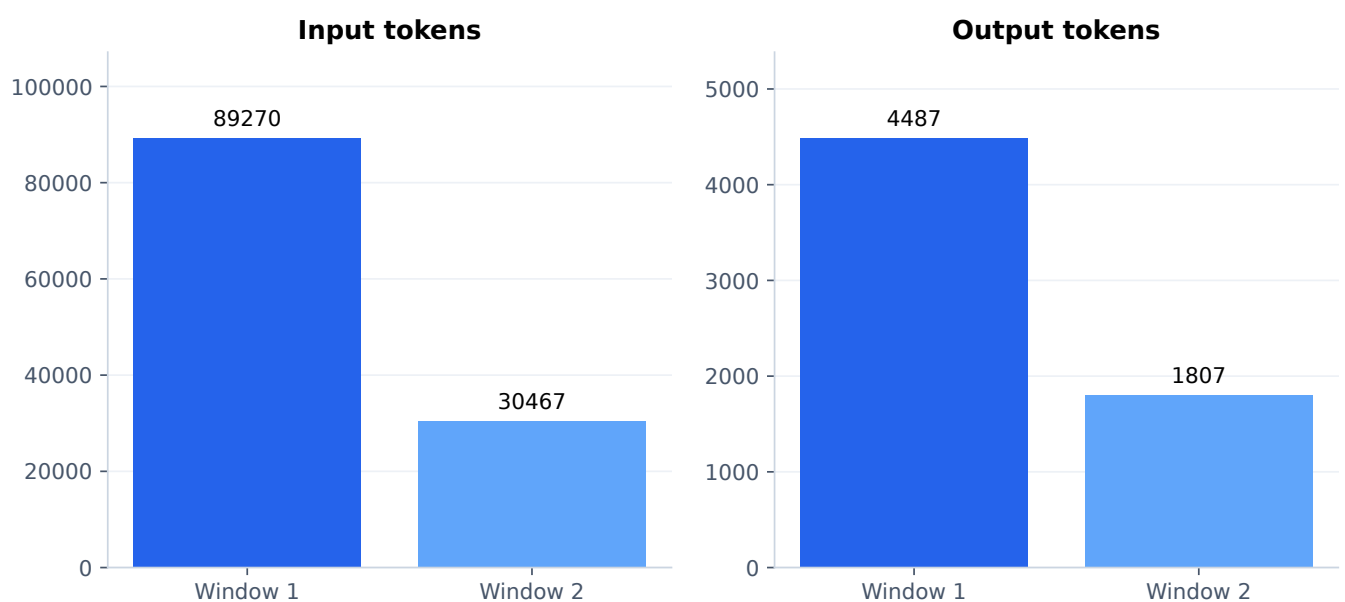
Fresh build latency per window



All six snapshots were built fresh. S3 includes real model maintenance; S0 and S1 are deterministic local writers. The log axis keeps small local fold times visible without implying they have equal cost.

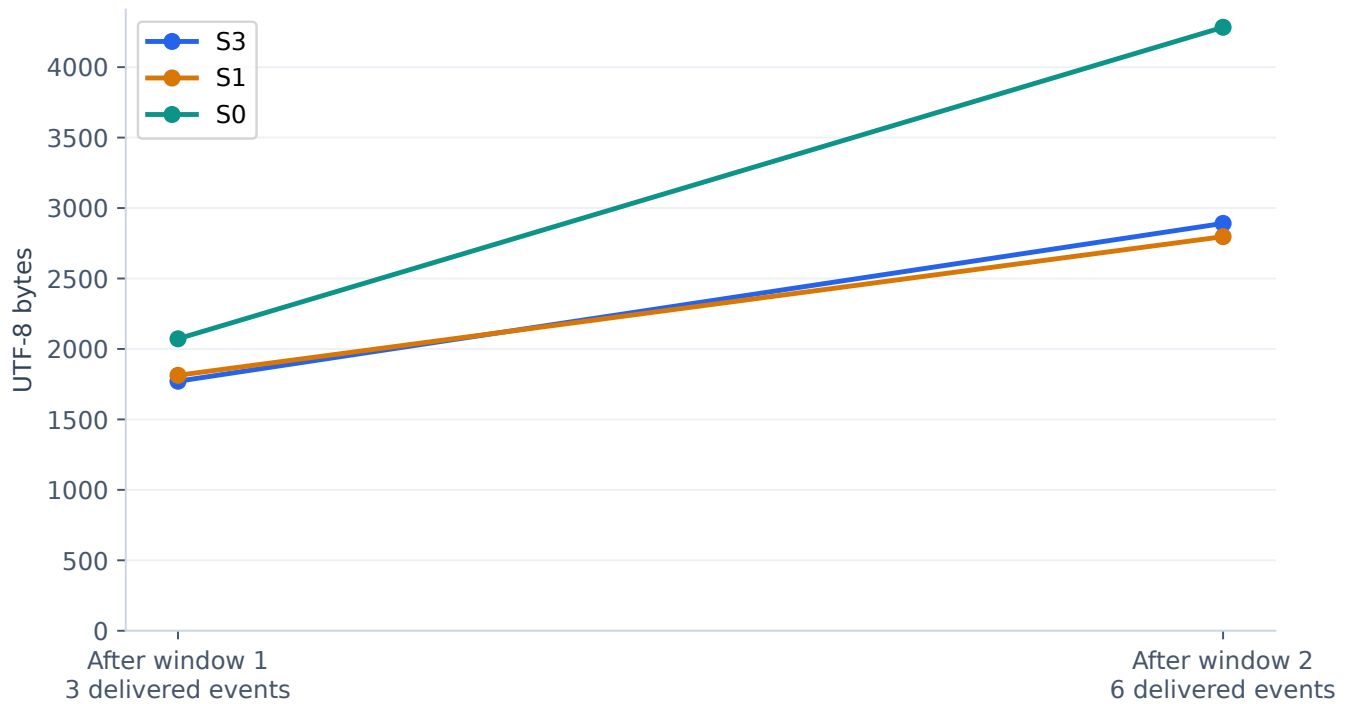
16

S3 builder token work



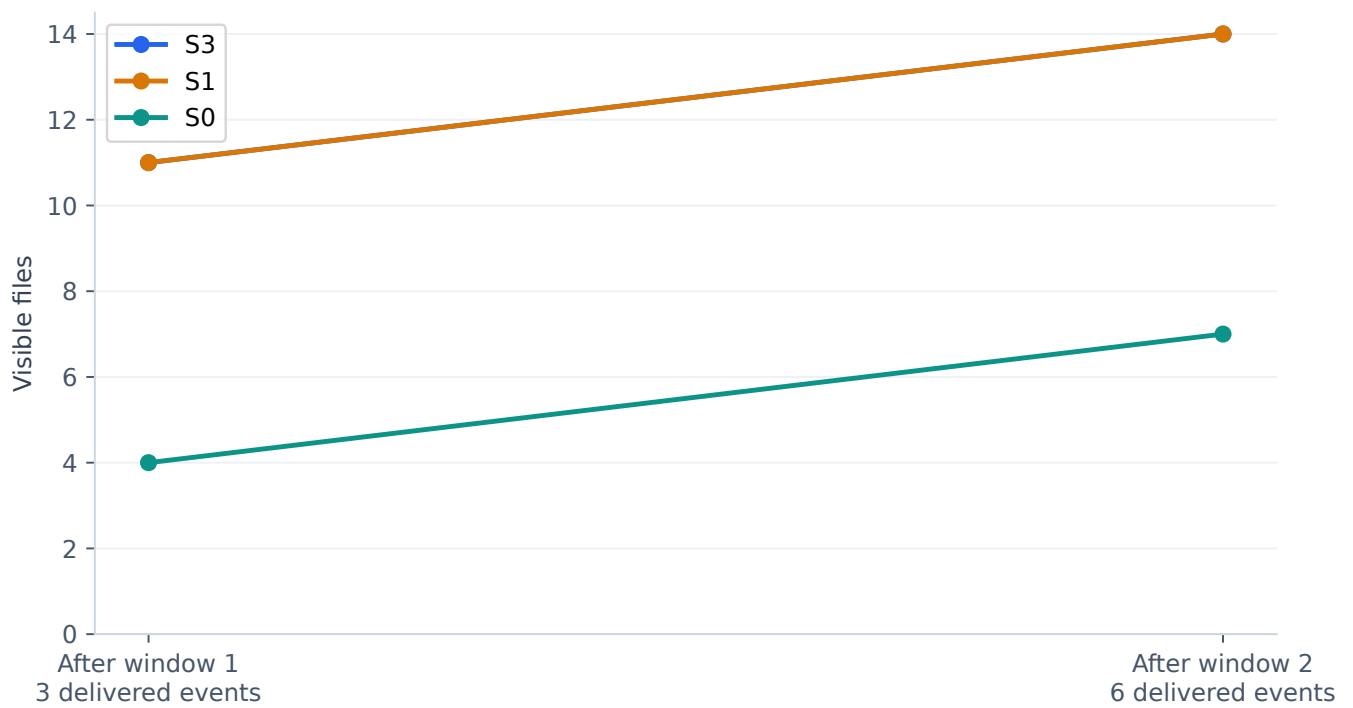
Each bar aggregates the Codex completions required for one successful fold. Deterministic S0/S1 writers make no builder model calls. These are new builds, not reused snapshots.

Reader-visible store growth



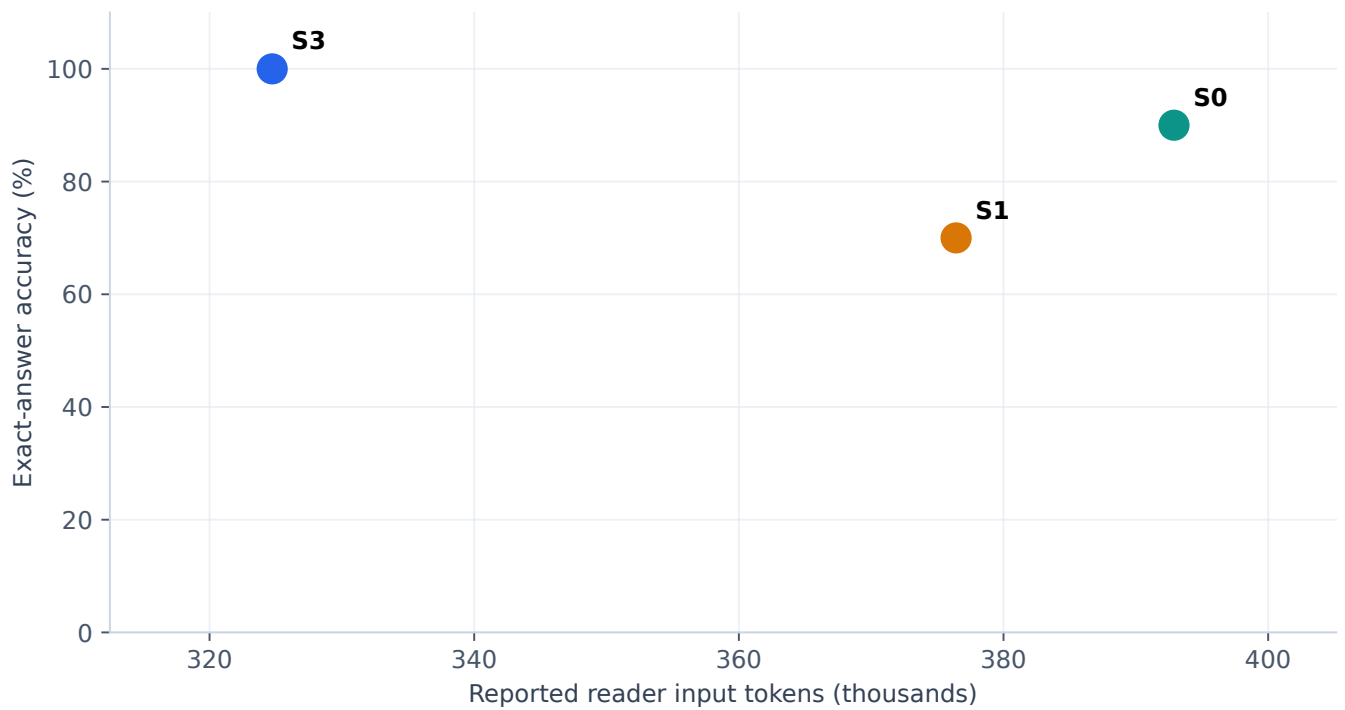
Measured from the exact reader-visible text export for each committed SHA. Git objects, evaluator ledgers, builder instructions and transient source mounts are excluded. Size alone is not semantic retention.

Reader-visible file count



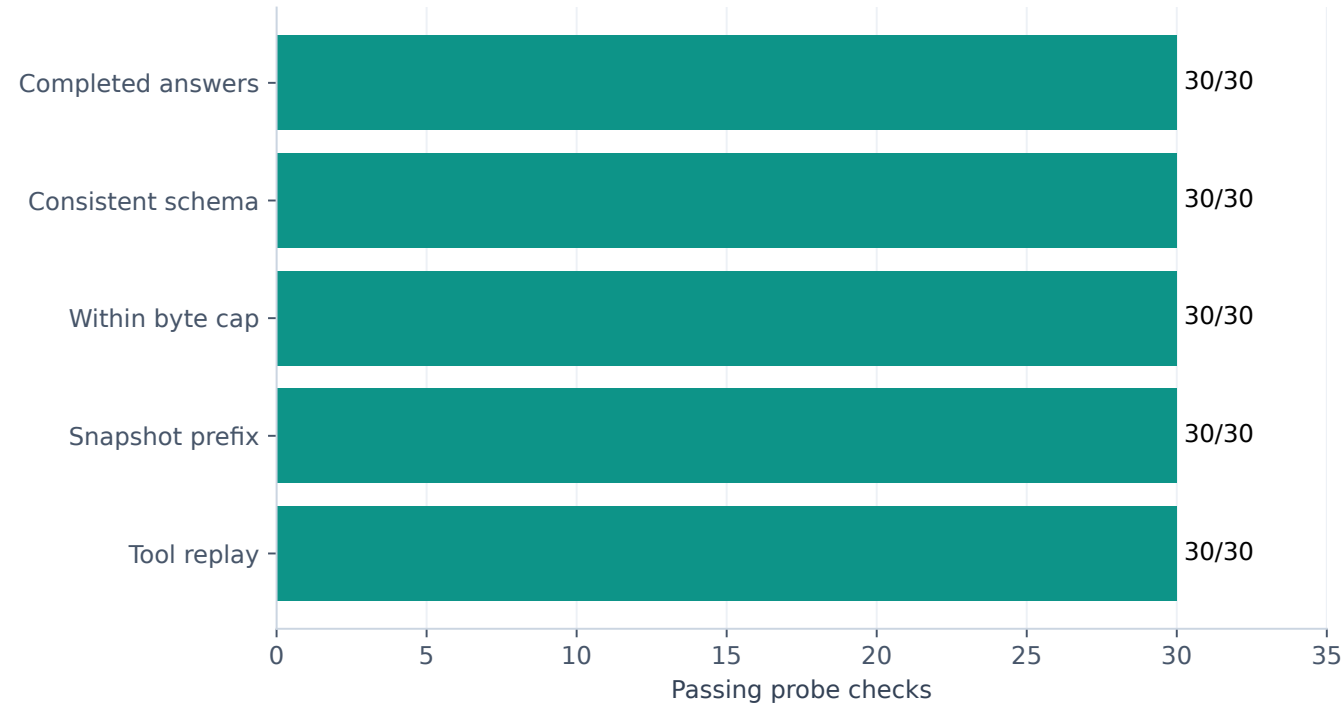
Measured from the exact reader-visible text export for each committed SHA. Git objects, evaluator ledgers, builder instructions and transient source mounts are excluded. Size alone is not semantic retention.

Observed quality versus input footprint



One point per store on one ten-probe workload. This is not a cost-optimal frontier: USD prices, independent build repetitions and held-out measurements are absent.

Infrastructure verification



These gates verify execution, access and accounting. They do not require correct answers. Tool replay recomputes every returned string from the selected SHA, including byte truncation.

The full question ledger

No wrong answers are removed. Open any row for its question, gold value, returned value, required and cited IDs, exact snapshot, and tool sequence. The linked JSON contains every raw tool return.

PROBE	STORE	EXACT VALUE	EVIDENCE	RETRIEVAL / RUNTIME	INSPECT
Q01 extraction	S3	Correct Mira	Pass	1,091 B 3 tools · 4 calls 22.1 s	▶ Question & trace
Q02 extraction	S3	Correct P1	Pass	1,880 B 2 tools · 3 calls 14.0 s	▶ Question & trace
Q03 temporal	S3	Correct Open	Pass	475 B 2 tools · 3 calls 15.7 s	▶ Question & trace
Q04 temporal	S3	Correct 2026-10-01	Pass	759 B 2 tools · 3 calls 20.3 s	▶ Question & trace

PROBE	STORE	EXACT VALUE	EVIDENCE	RETRIEVAL / RUNTIME	INSPECT
Q05 update	S3	Correct Noah	Pass	674 B 2 tools · 3 calls 15.2 s	► Question & trace
Q06 update	S3	Correct 2026-10-08	Pass	322 B 1 tools · 2 calls 10.7 s	► Question & trace
Q07 multisource	S3	Correct ["PR-71","PR-72"]	Pass	2,023 B 3 tools · 4 calls 21.7 s	► Question & trace
Q08 multisource	S3	Correct ["src/cache.py","tests/test_cache.py"]	Pass	2,407 B 4 tools · 5 calls 29.1 s	► Question & trace
Q09 abstention	S3	Correct unknown	Pass	2,322 B 5 tools · 6 calls 34.3 s	► Question & trace
Q10 abstention	S3	Correct unknown	Fail	888 B 5 tools · 6 calls 40.0 s	► Question & trace
Q01 extraction	S1	Correct Mira	Pass	606 B 2 tools · 3 calls 13.9 s	► Question & trace
Q02 extraction	S1	Correct P1	Pass	1,280 B 2 tools · 3 calls 15.3 s	► Question & trace
Q03 temporal	S1	Correct Open	Pass	475 B 2 tools · 3 calls 15.2 s	► Question & trace
Q04 temporal	S1	Incorrect unknown	Fail	532 B 5 tools · 6 calls 40.3 s	► Question & trace
Q05 update	S1	Correct Noah	Pass	1,280 B 2 tools · 3 calls 14.8 s	► Question & trace
Q06 update	S1	Incorrect unknown	Pass	698 B 5 tools · 6 calls 35.0 s	► Question & trace
Q07 multisource	S1	Correct ["PR-71","PR-72"]	Pass	2,000 B 5 tools · 6 calls 36.4 s	► Question & trace
Q08 multisource	S1	Incorrect unknown	Fail	1,819 B 5 tools · 6 calls 37.6 s	► Question & trace
Q09 abstention	S1	Correct unknown	Pass	1,280 B 4 tools · 5 calls 27.7 s	► Question & trace
Q10 abstention	S1	Correct unknown	Fail	732 B 5 tools · 6 calls 32.2 s	► Question & trace

PROBE	STORE	EXACT VALUE	EVIDENCE	RETRIEVAL / RUNTIME	INSPECT
Q01 extraction	S0	Correct Mira	Pass	809 B 3 tools · 4 calls 20.1 s	▶ Question & trace
Q02 extraction	S0	Correct P1	Pass	1,519 B 3 tools · 4 calls 23.9 s	▶ Question & trace
Q03 temporal	S0	Correct Open	Pass	1,453 B 3 tools · 4 calls 23.6 s	▶ Question & trace
Q04 temporal	S0	Correct 2026-10-01	Pass	765 B 3 tools · 4 calls 20.8 s	▶ Question & trace
Q05 update	S0	Correct Noah	Pass	1,695 B 4 tools · 5 calls 25.5 s	▶ Question & trace
Q06 update	S0	Correct 2026-10-08	Pass	1,092 B 4 tools · 5 calls 26.4 s	▶ Question & trace
Q07 multisource	S0	Correct ["PR-71","PR-72"]	Pass	3,049 B 5 tools · 6 calls 35.9 s	▶ Question & trace
Q08 multisource	S0	Incorrect unknown	Fail	2,692 B 5 tools · 6 calls 36.4 s	▶ Question & trace
Q09 abstention	S0	Correct unknown	Fail	1,056 B 5 tools · 6 calls 33.8 s	▶ Question & trace
Q10 abstention	S0	Correct unknown	Pass	1,581 B 5 tools · 6 calls 34.7 s	▶ Question & trace

Exactly what ran

Model gpt-5.6-luna

Reasoning effort medium

Codex CLI codex-cli 0.154.0

Builder Harnext + Codex host-controlled file tools

Reader Codex JSON actions; host list / read / search

Workload	6 constructed events; 10 probes × 3 stores
Windows	Two hourly event-time windows; 3 events in each
Retrieval allowance	32,768 UTF-8 bytes per probe; 12,000 per tool response
Reader allowance	6 completions; final slot restricted to an answer
Builder allowance	40 action completions; 180 s per fold
Replication	1 fresh store build per arm; 1 reader attempt per probe
Execution	3 arm processes concurrently; serial questions within each arm

This is the initial merged E2/E3 path. The native Codex shell comparison remains unavailable on this host because its bwrap sandbox failed to initialize in the earlier trial. This run explicitly uses host-controlled file tools; it does not bypass the native sandbox.

Measured usage in this fresh run

Builder: **119,737** reported input tokens and **6,294** output tokens. Readers: **1,094,034** input and **12,391** output. Cache hits are included within input totals. S0/S1 use deterministic writers without builder model calls. USD billing is unknown.

The retrieval cap is **32,768 UTF-8 bytes**, not 32k model tokens or a verified 8k-token context budget. Provider input also includes fixed instructions and repeated conversation history. The output-token target is advisory. USD billing is unknown; the token charts report measured usage.

Same delivered prefixes, separate store snapshots

STORE	FIRST SNAPSHOT	SECOND SNAPSHOT	DELIVERED EVENTS
S3	cb2342b11ac21131d4ffa79bf34bd33c c1a5617d	256dcacdaeacf8b4e479ad91e37ea63f e1be427c	3 → 6
S1	c2bf11566d962e8e8f45483212f8e347 4442e4e9	84db2cbd0225880b8b53e1409ccdf0f7 7df131f6	3 → 6
S0	dfd2e4ad1fac09b180d8dadd651152b4 f975882f	1e926611ca2ad9ee556d8cd104169e10 bf08bd50	3 → 6

Which study graphs cannot be made from this smoke?

This report includes 20 plots computed from the measured smoke records. The full blueprint calls for additional experiments; drawing their curves now would invent observations.

PLANNED COMPARISON	WHAT IS MISSING
Accuracy versus 2k / 8k / 32k token budgets	A pinned tokenizer and multiple executed token-budget conditions. This run uses one byte cap.
File-template, preservation-policy and tool-policy interactions	The full declared file-condition matrix; current S3 and S1 are individual starting configurations.
BM25, dense retrieval, reranking and graph-hop curves	Those retrieval/graph conditions and their relevance/edge ground truth were not executed here.
Harness or model comparisons	Only Codex Luna medium with file tools was run. Concurrent arms are stores, not different harnesses.
Confidence intervals, significance and robustness	Enough independent store builds and representative workload clusters; two tiny smoke observations are insufficient.
USD cost frontier, amortization and long-stream scaling	Frozen billing prices, reuse workloads and longer streams. Reported token counts and two-window sizes are shown instead.
Classifier routing, Kafka throughput and freshness under load	These are outside this isolated context-quality smoke; no E1 or transport result is implied.

Evidence and shareable exports

S3 evidence

- [Manifest & source hashes](#)
- [Exact configuration](#)
- [Original scored results](#)

S1 evidence

- [Manifest & source hashes](#)
- [Exact configuration](#)
- [Original scored results](#)

S0 evidence

- [Manifest & source hashes](#)
- [Exact configuration](#)
- [Original scored results](#)

[Independent tool replay
audit](#)

[Reader-visible snapshot
contents](#)

[Frozen probes and gold](#)

[Frozen event replay](#)

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[Frozen event replay](#)

Model prompts, budgets, gold and scoring were not tuned during this rerun. All six snapshots were built fresh. Report generation and deterministic auditing are read-only with respect to the model outputs and original scores.

► [Frozen inputs and source location](#)

E2 context-engine smoke · 12 September 2026. Self-contained HTML: no external fonts, scripts, trackers or chart libraries. Figures remain visible without the sidecar folder; local evidence/download links require the accompanying assets. Open the ZIP to keep the report and exports together. This report records a synthetic infrastructure check, not a thesis confirmation result.